

GNN_MODEL V2

Model: "GNNModel"

Layer (type)	Output Shape	Param #	Connected to
model_input (InputLayer)	[(None, 16, 20)]	0	[]
bn_input (BatchNormalization) ['model_input[0][0]']	(None, 16, 20)	80	
adj_input (InputLayer)	[(None, 16, 16)]	0	[]
gcn_dense_0 (QDense) ['bn_input[0][0]']	(None, 16, 32)	672	
gcn_agg_0 (Dot) ['adj_input[0][0]', 'gcn_dense_0[0][0]']	(None, 16, 32)	0	
gcn_act_0 (QActivation) ['gcn_agg_0[0][0]']	(None, 16, 32)	0	
gcn_dense_1 (QDense) ['gcn_act_0[0][0]']	(None, 16, 16)	528	
gcn_agg_1 (Dot) ['adj_input[0][0]', 'gcn_dense_1[0][0]']	(None, 16, 16)	0	
gcn_act_1 (QActivation) ['gcn_agg_1[0][0]']	(None, 16, 16)	0	
act_pool (QActivation) ['gcn_act_1[0][0]']	(None, 16, 16)	0	
pool (GlobalMaxPooling1D) ['act_pool[0][0]']	(None, 16)	0	
dense_0 (QDense) [0]']	(None, 16)	272	['pool[0]
act_0 (QActivation) [0]']	(None, 16)	0	['dense_0[0]

jet_pt_reg_0 (QDense)	(None, 8)	136	['act_0[0][0]']
jet_id_classification (QDense)	(None, 8)	136	['act_0[0][0]']
jet_pt_reg_act_0 (QActivation)	(None, 8)	0	
jet_id_output (Softmax)	(None, 8)	0	
pT_output (QDense)	(None, 1)	9	

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Total params: 1833 (7.16 KB)
Trainable params: 1793 (7.00 KB)
Non-trainable params: 40 (160.00 Byte)
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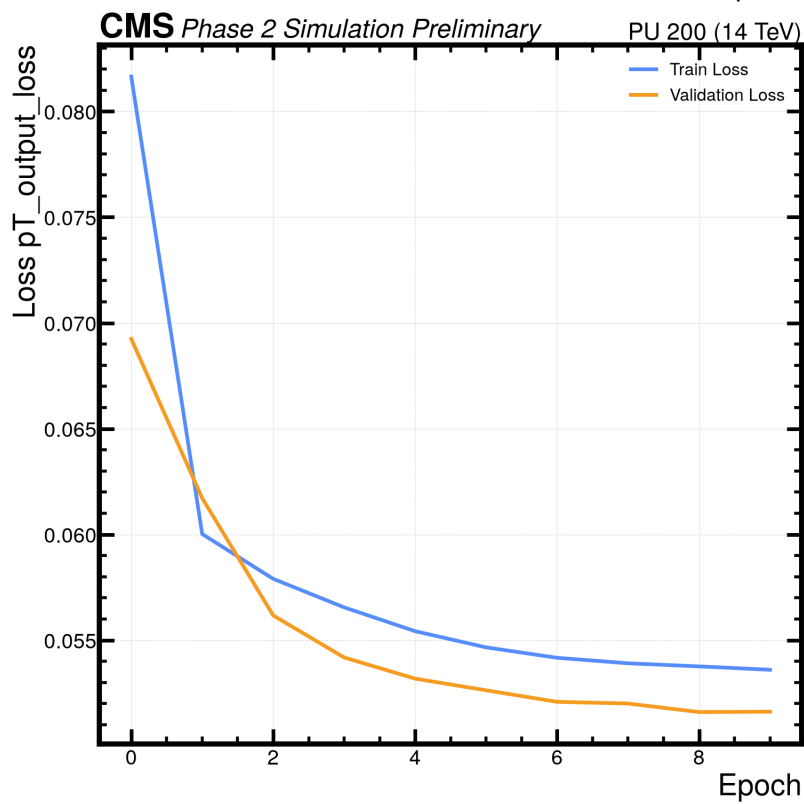
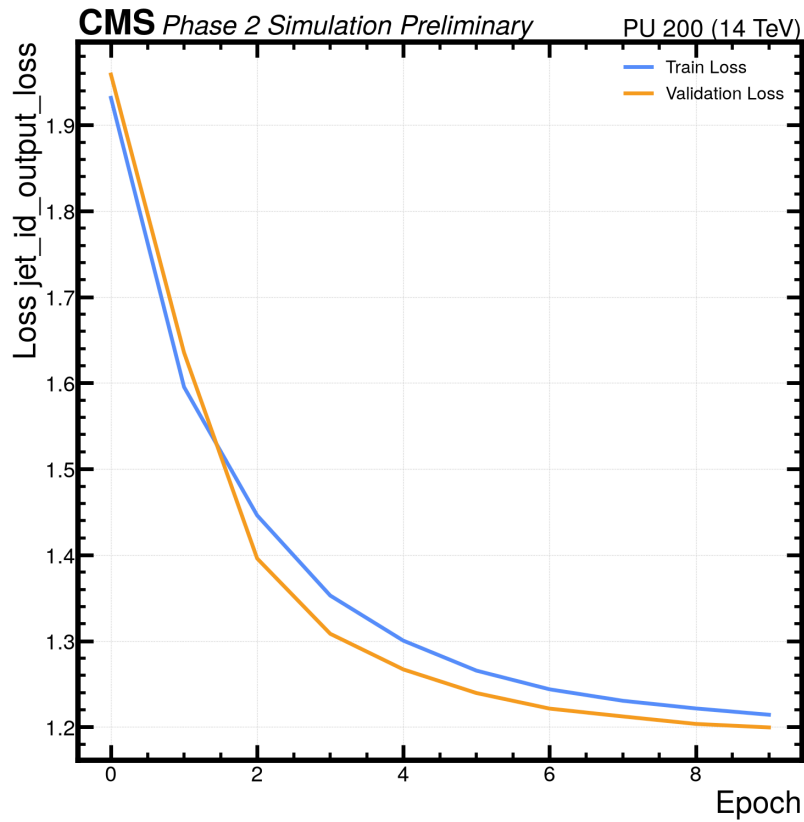
```

Epoch 10/10
153/153 - 2s -
loss: 1.2679 -
jet_id_output_loss: 1.2143 -
pT_output_loss: 0.0536 -
jet_id_output_categorical_accuracy: 0.5098 -
jet_id_output_weighted_categorical_accuracy: 0.5326 -
pT_output_mae: 0.2471 -
pT_output_mean_squared_error: 0.1127 -
pT_output_weighted_mae: 0.2359 -
pT_output_weighted_mean_squared_error: 0.1072 -
val_loss: 1.2511 -
val_jet_id_output_loss: 1.1995 -
val_pT_output_loss: 0.0516 -
val_jet_id_output_categorical_accuracy: 0.5205 -
val_jet_id_output_weighted_categorical_accuracy: 0.5381 -
val_pT_output_mae: 0.2480 -
val_pT_output_mean_squared_error: 0.1115 -
val_pT_output_weighted_mae: 0.2327 -
val_pT_output_weighted_mean_squared_error: 0.1035 -
lr: 0.0010

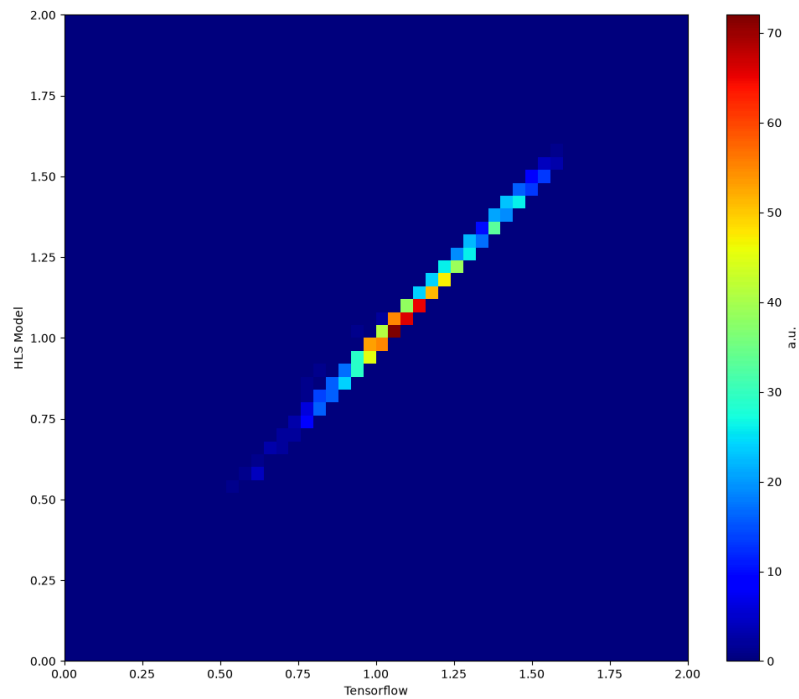
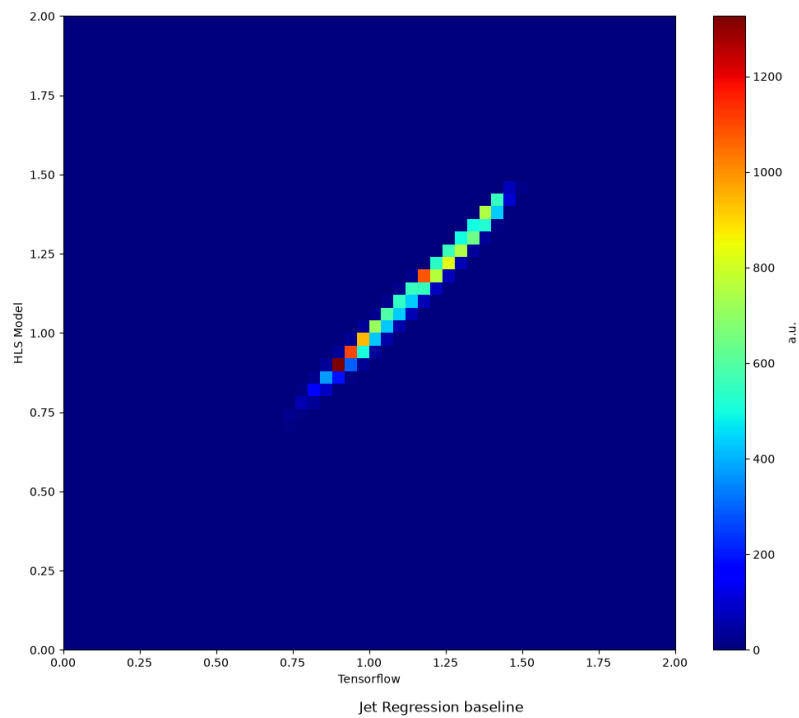
```

Baseline Deepset Model params:
Total params: 2776 (10.84 KB)
Trainable params: 2736 (10.69 KB)
Non-trainable params: 40 (160.00 Byte)

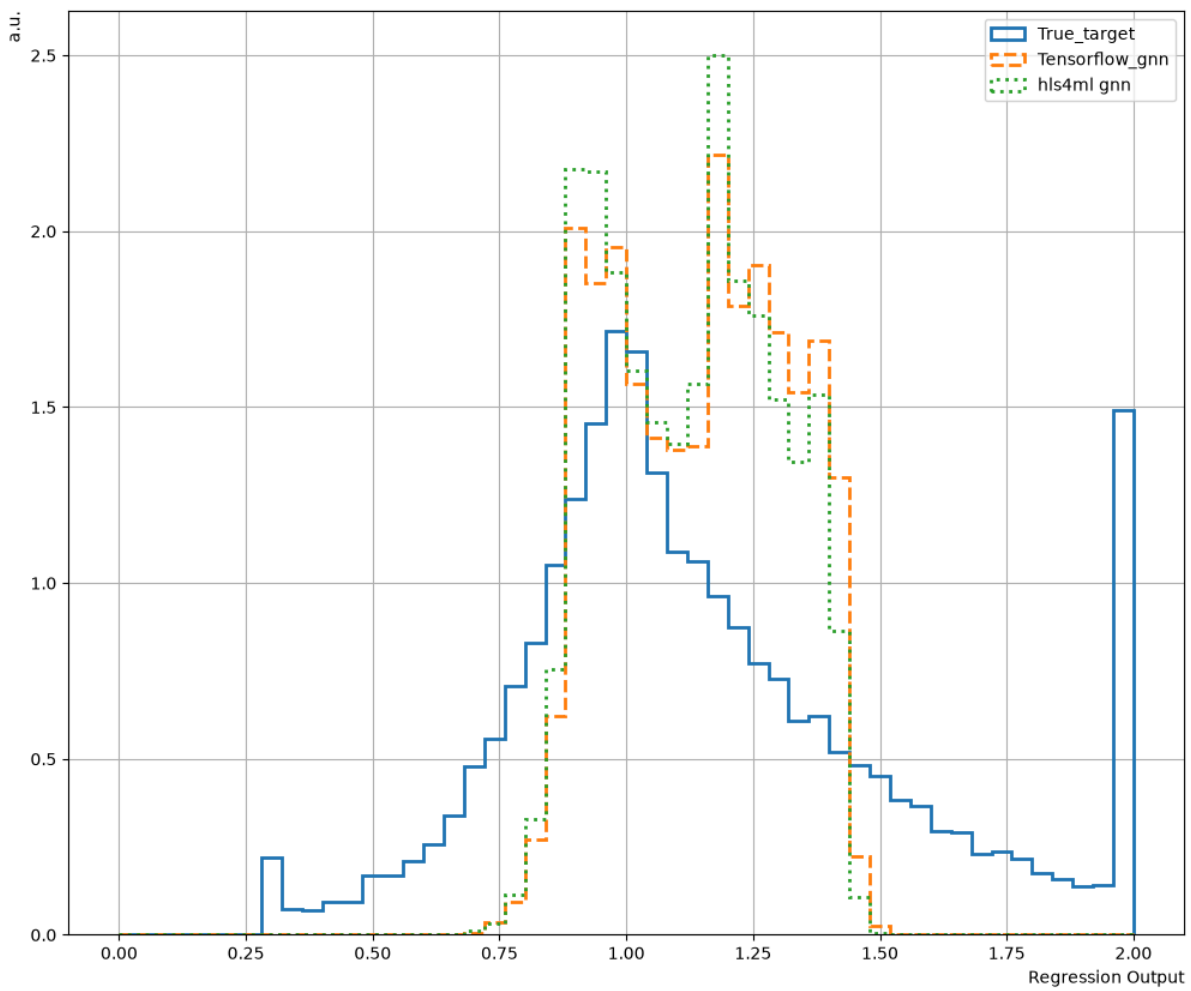
Loss Curves for GNN model



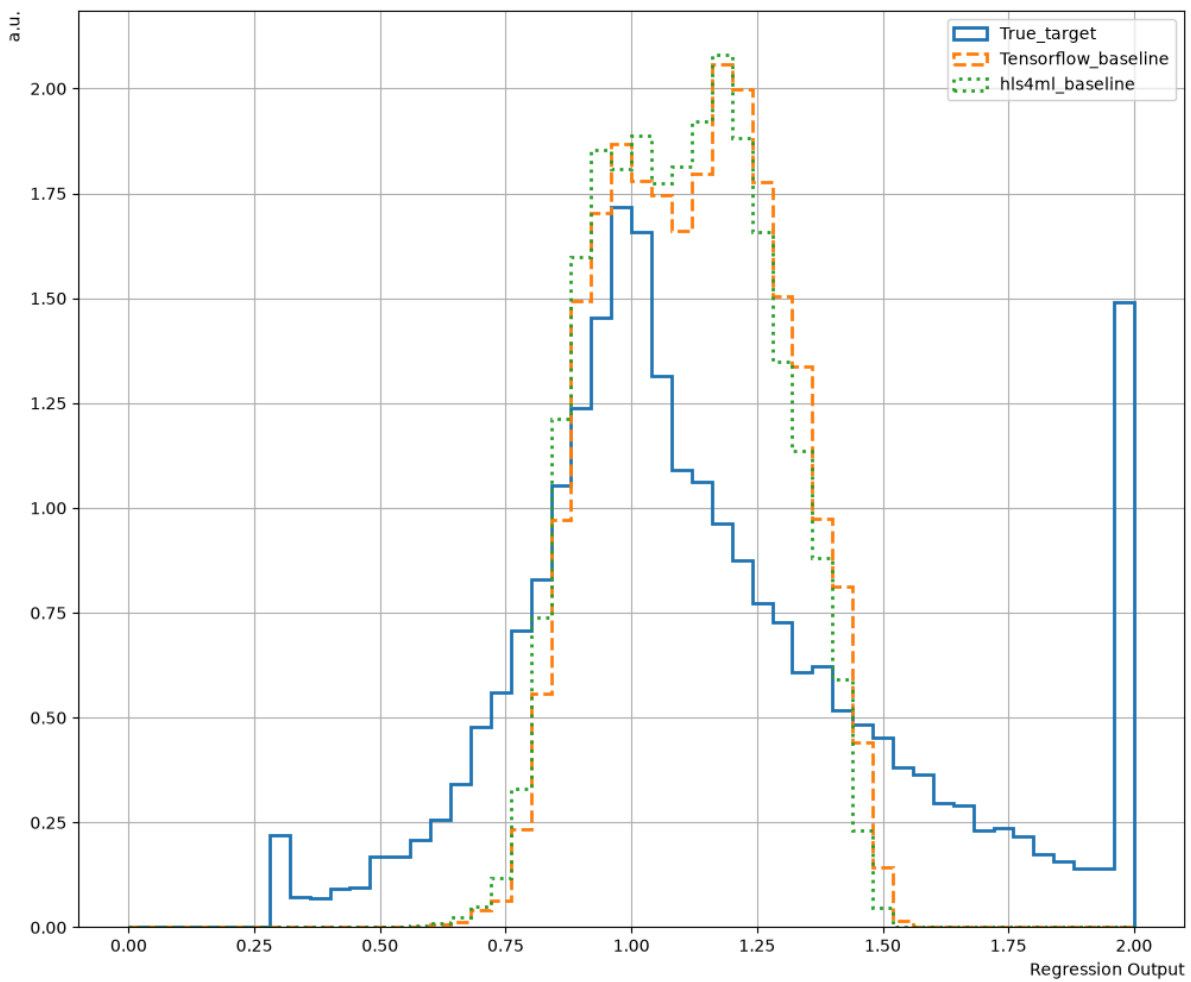
Regression Head Overlap comparison GNN vs Baseline



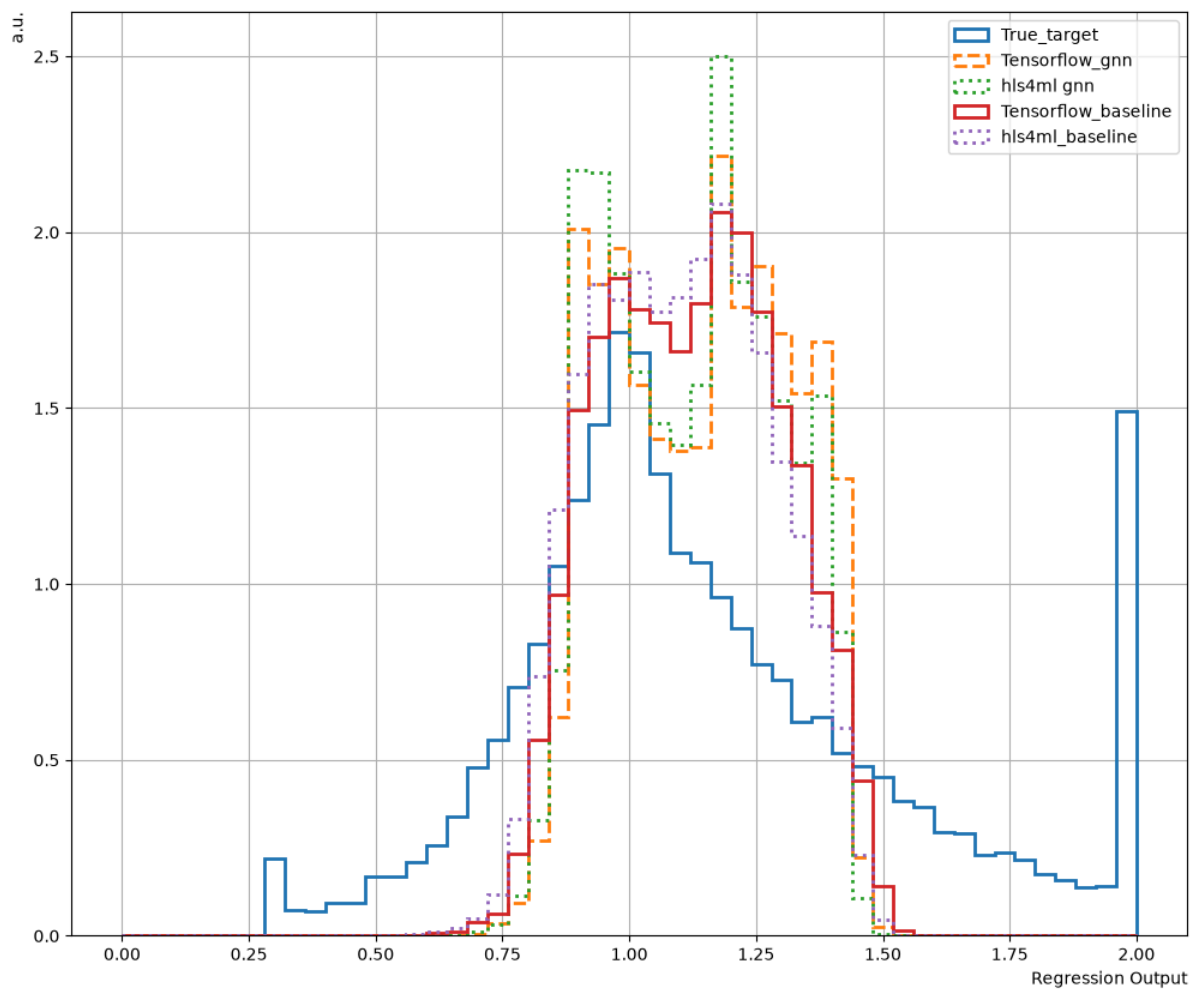
True Regression output vs GNN model



True Regression output vs Baseline model



True Regression output vs All Models



Classification confusion matrix for all Models:

--- GNN (TensorFlow) ---

Overall accuracy: 0.5203

--- GNN (hls4ml) ---

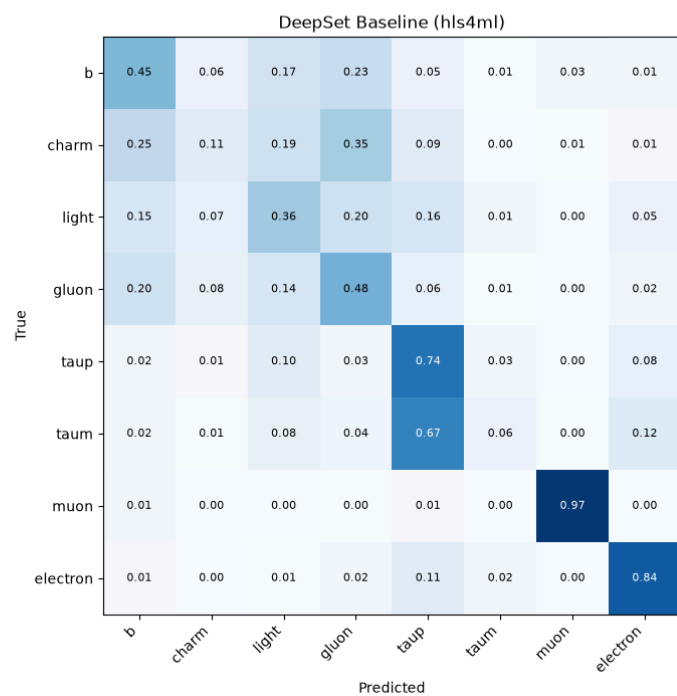
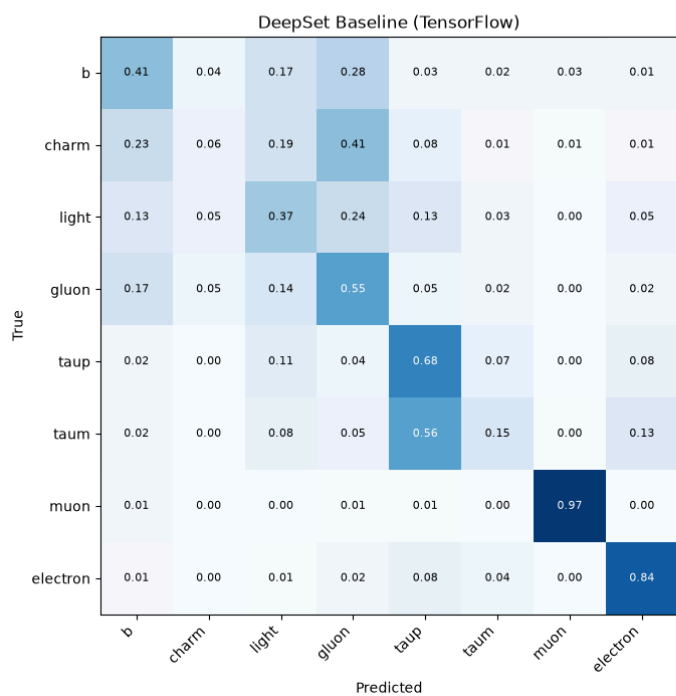
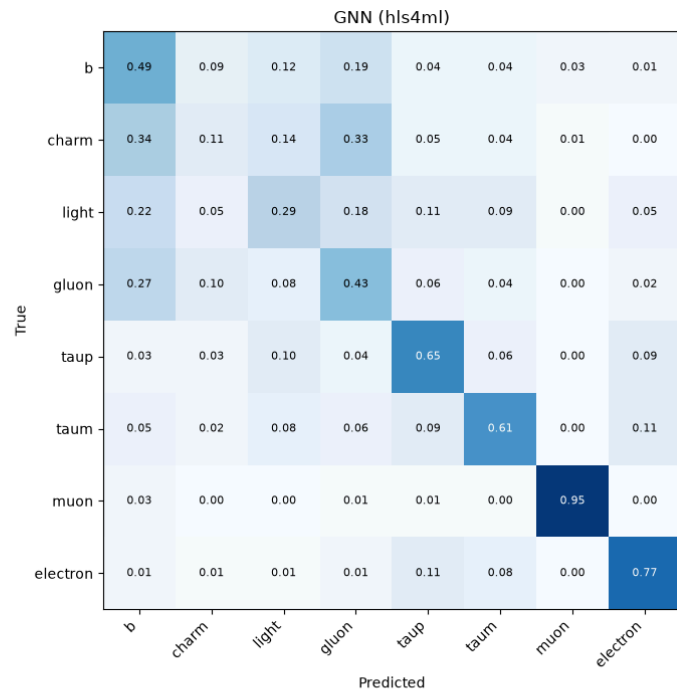
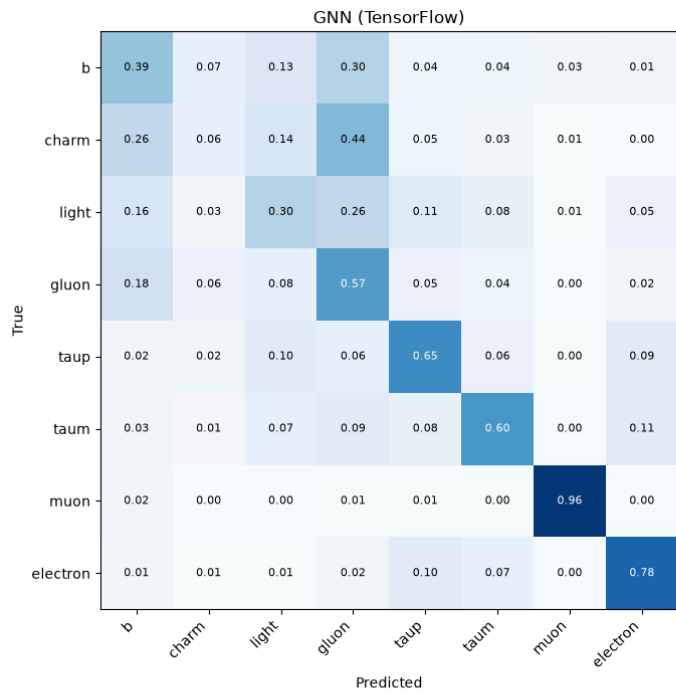
Overall accuracy: 0.5331

--- DeepSet Baseline (TensorFlow) ---

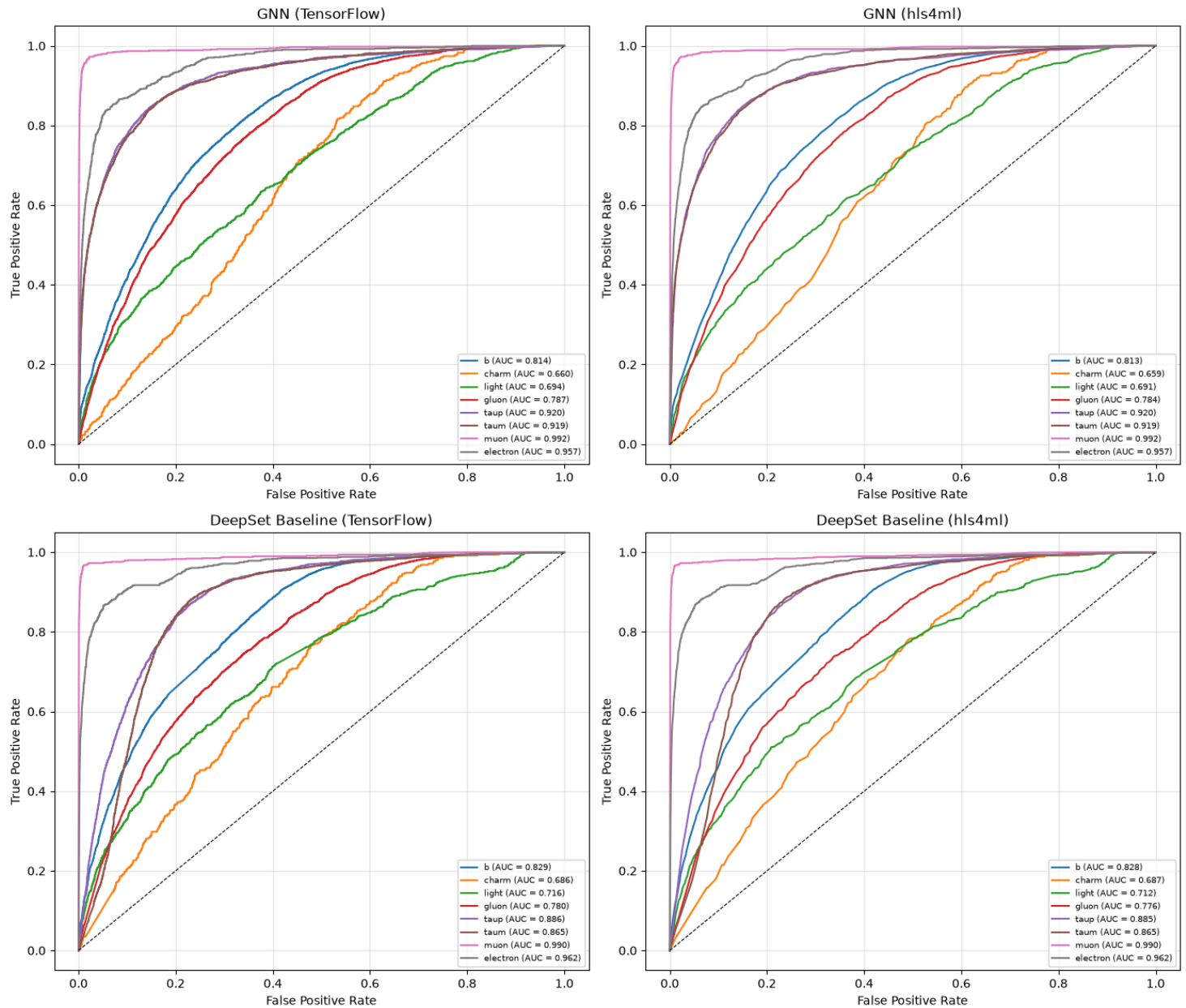
Overall accuracy: 0.4800

--- DeepSet Baseline (hls4ml) ---

Overall accuracy: 0.4743



Classification ROC Curves for all models



Classification numbers:

Test shape: (19230, 8)

--- GNN (TensorFlow) ---

Class	Recall (per-class acc)	OvR accuracy	Support
b	0.3852	0.7059	7477
charm	0.0630	0.9413	349
light	0.3044	0.8748	992
gluon	0.5728	0.7607	3670
taup	0.6474	0.9105	2280
taum	0.6028	0.9126	2457
muon	0.9575	0.9840	989
electron	0.7835	0.9509	1016

Overall accuracy: 0.5203

--- GNN (hls4ml) ---

Class	Recall (per-class acc)	OvR accuracy	Support
b	0.4902	0.7208	7477
charm	0.1060	0.9222	349
light	0.2853	0.8772	992

gluon	0.4270	0.7882	3670
taup	0.6487	0.9096	2280
taum	0.6068	0.9110	2457
muon	0.9545	0.9858	989
electron	0.7736	0.9515	1016

Overall accuracy: 0.5331

--- DeepSet Baseline (TensorFlow) ---

Class	Recall (per-class acc)	OvR accuracy	Support
b	0.4139	0.7230	7477
charm	0.0602	0.9562	349
light	0.3669	0.8461	992
gluon	0.5540	0.7722	3670
taup	0.6763	0.8558	2280
taum	0.1506	0.8697	2457
muon	0.9676	0.9852	989
electron	0.8356	0.9519	1016

Overall accuracy: 0.4800

--- DeepSet Baseline (hls4ml) ---

Class	Recall (per-class acc)	OvR accuracy	Support
b	0.4471	0.7292	7477
charm	0.1060	0.9405	349
light	0.3599	0.8476	992
gluon	0.4782	0.7860	3670
taup	0.7386	0.8358	2280
taum	0.0562	0.8701	2457
muon	0.9676	0.9855	989
electron	0.8356	0.9538	1016

Overall accuracy: 0.4743